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Introduction to Cryptography

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Cryptography is nothing more than a mathematical framework for discussing the implications of various paranoid delusions.

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Agenda

1. Introduction
2. Cryptographic Hash (one-way) functions & MACs
3. Symmetric key cryptography
4. Public key cryptography
5. Digital signatures
6. Authentication – key establishment

Cryptography

Cryptology: cryptography and cryptanalysis

- **Cryptography** is the science and study of secret writing
- **Cryptanalysis** is the science and study of methods of breaking ciphers

Today:

- cryptography is the study of mathematical techniques related to aspects of information security, such as confidentiality, data integrity, entity authentication, and data origin authentication

Note:

Cryptography does not solve security problems; it just transforms them into other problems, which are hopefully easier to handle.

The origins of cryptography





The enemy is an outsider listening to traffic

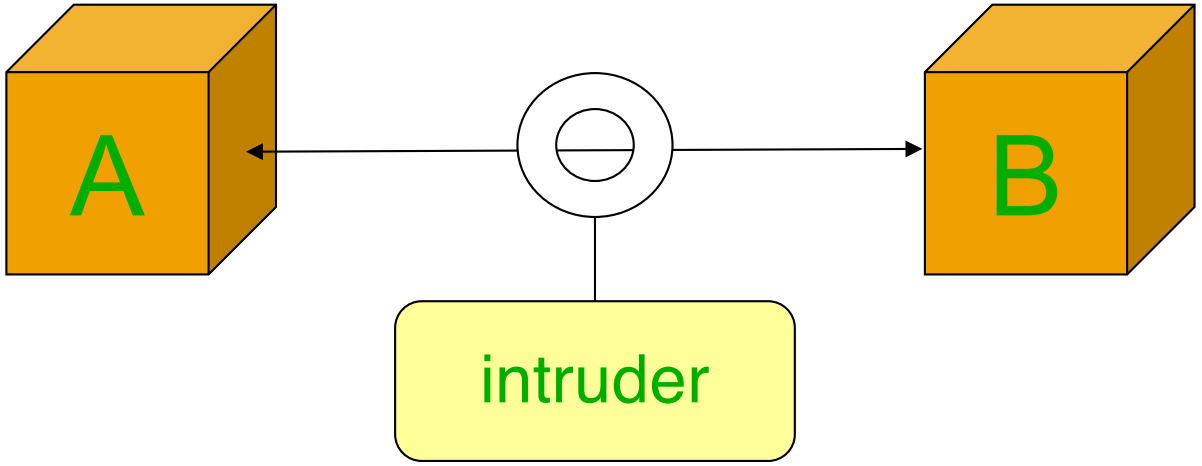
Two secure end systems communicate over an insecure channel



How do we turn untrustworthy channels into trustworthy ones?

- **Cryptography is an enabling technology.**

The old paradigm



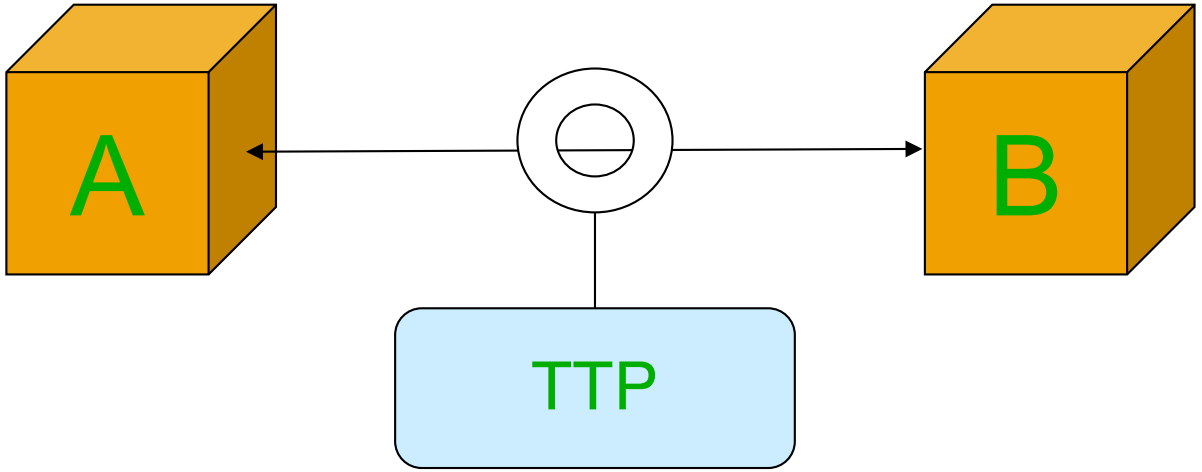
A **and** ***B*** **communicate over an insecure channel.**

A **and** ***B*** **trust each other.**

Intruder can read, delete, and insert messages.

With cryptography, ***A*** **and** ***B*** **construct a secure logical channel over an insecure network.**

The new paradigm

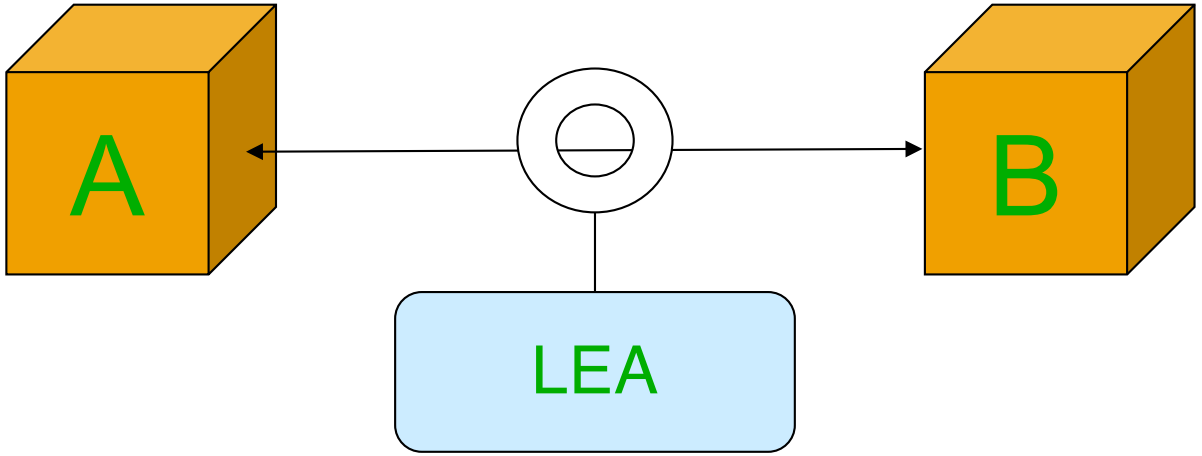


Electronic commerce: *A* and *B* are customer and merchant; they do not "trust" each other.

We want protection against insider fraud as much as protection against outsiders.

Trusted Third Parties help settle disputes.

Law enforcement

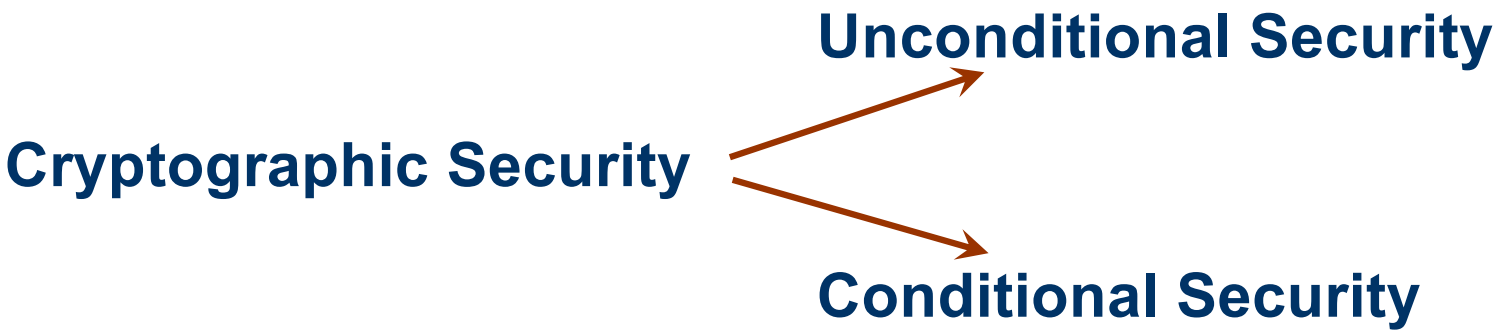


In many countries laws regulate how a law enforcement agency (LEA) can intercept traffic.

Key recovery makes cryptographic keys available to their owner.

Key escrow makes keys available to a LEA.

Cryptographic security



- **Unconditional Security: System is secure even if adversary has unbounded computing power.**
  - **Security can be measured using information theory**

- **Conditional Security: System can be broken in principle, but this requires more computing power than a realistic adversary would have.**
  - **Security can be measured using complexity theory**

Cryptographic security services

Data confidentiality: encryption hides the content of messages

- Goal: Only authorized users get the content

Data integrity: integrity check functions (hash functions) detect changes to data

- Goal: Verify that data is correct & complete, or detect that this is not the case

Data origin authentication: digital signatures and message authentication codes verify the source and integrity of data

- Goal: Verify that data is original (not modified) and comes from a specific source, or detect that this is not the case

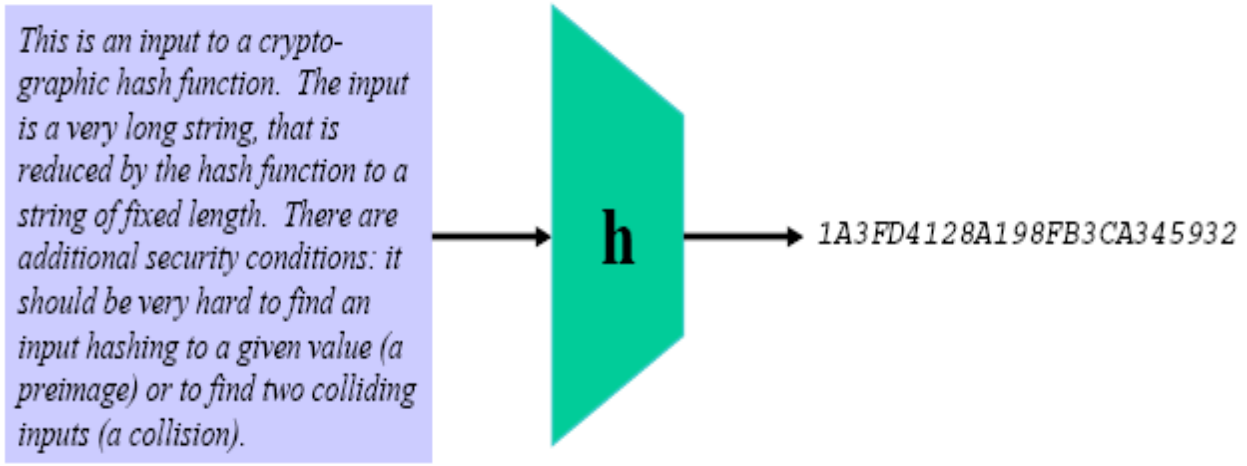
Cryptographic hash functions

Cryptographic hash functions are often used for integrity checks

Apply a hash function *h* to a document *x* and store the result *h(x)* in a secure place

The result *h(x)* is called "hash value", "message digest", or "checksum"

Changes to *x* can be detected by re-computing the hash of *x* and comparing the result with the stored value



Definition of *hash functions* (see HAC)

A *hash function* is a function h which as, as a minimum, the following two properties:

Ease of computation: Given h and an input x, it is easy to compute *h(x)*

Compression: the hash function maps inputs of arbitrary length to fixed length results

Note: *hash function* usually implies an unkeyed hash function.
a *keyed hash function* is a function *h(x,k)* that takes some arbitrary-length input *x* and a fixed-length key *k* as input

Functional classification of cryptographic hash functions

Manipulation detection codes (MDCs)
(also: *modification detection code* or *message integrity code*):

Used to detect changes to a document; unkeyed hash

Two types of MDCs:

- **One-way hash function (OWHF): finding an input which hashes to a pre-specified hash-value is difficult**
- **Collision resistant hash function (CRHF): finding any two inputs having the same hash value is difficult**

Message authentication codes (MACs):

Used to assure both the source of a message and its integrity; keyed hash function

- **see later**

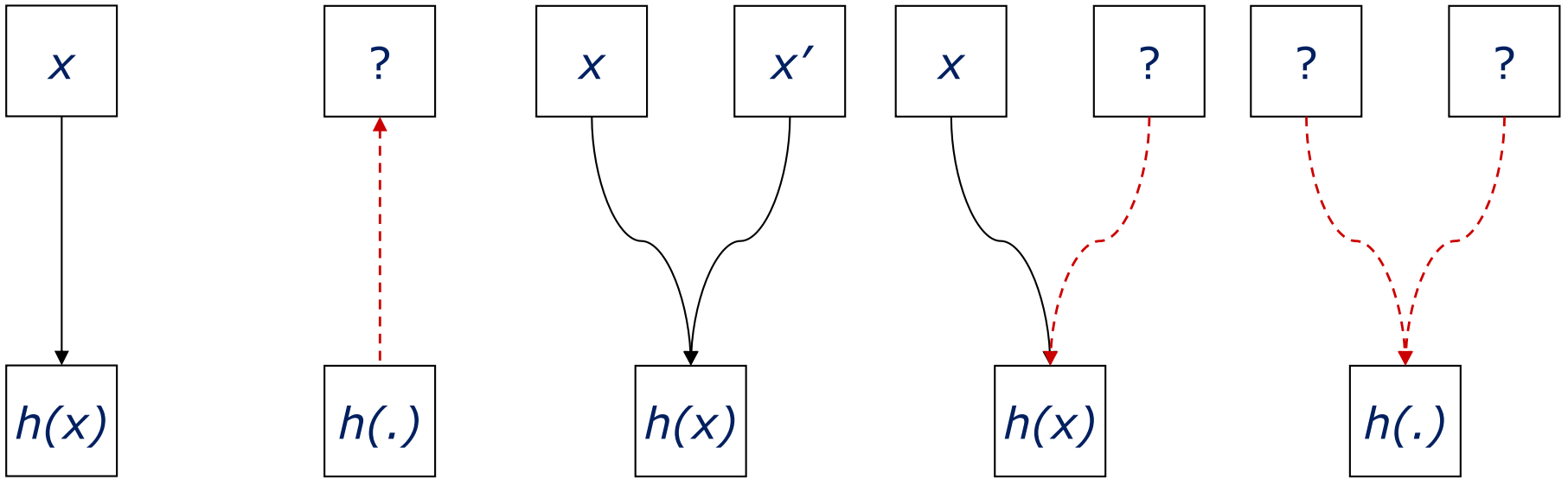
Security properties of hash functions (see HAC)

Pre-image resistance (one-way): for any given y, it is computationally infeasible to find x so that $h(x)=y$

2nd pre-image resistance: given x, it is computationally infeasible to find x', $x \neq x'$, so that $h(x)=h(x')$

Collision resistance: it is computationally infeasible to find x and x', $x \neq x'$, so that $h(x)=h(x')$

Properties of one-way functions



ease of computation

pre-image resistance

collision

2nd pre-image resistance

collision resistance

Hash functions

One-way hash function (OWHF)

- ease-of-computation, compression,
  pre-image resistance, 2nd-preimage resistance
- **One-way functions can e.g. be used for storing passwords**

Collision resistant hash function (CRHF)

- ease-of-computation, compression,
  2nd-preimage resistance, collision resistance
  - **Note 1: Collision resistance implies 2nd-preimage resistance**
  - **Note 2: In practice, CRHF usually are preimage resistant as well**
  - **Note 3: Some pathological examples are collision resistant but not preimage resistant**

Well known hash functions: SHA-1, RIPEMD-160, MD5